

Lab_08

Kenya Sanchez

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Kenya Sanchez

Section 1: Overview

radius: mean of distances from nucleus center to points on the perimeter;

texture: a measure of nucleus roughness taken from the standard deviation of gray-scale values;

perimeter: total boundary length of the nucleus,

area: total area of the nucleus;

smoothness: local variation in radius lengths, i.e. how “bumpy” the edge is;

compactness: measures how circular vs. irregular the shape is;

concavity: how deeply indented, and

symmetry: how symmetric the nucleus is.

Section 2: Exploratory data analysis

Getting organized

Question 11 and 15 are optional extensions They can be skipped

Preparing the data

Download and Import Data

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
```

Examine your input data to ensure column names are set correctly

```
# View the first 4 rows
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
84348301	0.14250	0.28390	0.2414		0.10520
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	

842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			

New Data Frame that Omits “Diagnosis” column

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[, -1]
```

Generate a vector for the “Diagnosis” data

```
# Create diagnosis vector for later
diagnosis <- factor(wisc.df$diagnosis)
```

Exploratory data analysis

Explore the data you created before (wisc.data and diagnosis) to answer the following questions:

Q1: How many observations are in this dataset?

Answer Q1: Each row represents a different observation and a different patient. Each column represents a variable that describes the observation made. In this dataset there are a total of 569 observation.

```
# #Observations = #Rows
dim(wisc.df)
```

```
[1] 569 31
```

```
nrow(wisc.df)
```

```
[1] 569
```

Q2: How many of the observations have a malignant diagnosis?

Answer Q2: There are a total of 212 observations with a malignant diagnosis

```
# Use table() function to reveal diagnosis of observations
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

```
# Call out Malignant diagnosis ONLY
table(diagnosis)["M"]
```

```

M
212
```

Q3: How many variables/features in the data are suffixed with __mean?

Answer Q3: There are 10 variables/features suffixed with “__mean” in this data set.

```
# How many variables / features present total
length(wisc.df)
```

```
[1] 31
```

```
# Specify variables with suffix __mean
length(grep("__mean", colnames(wisc.df)))
```

```
[1] 10
```

Section 3: Principal Component Analysis

Performing PCA

Check Mean and Standard Deviation of features

```
# Check column means and standard deviations  
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se

2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

Execute PRC

```
# Perform PCA on wisc.data by completing the following code
wisc.pr <- prcomp(wisc.data, scale.=TRUE)
```

Summary of Results

```
# Look at summary of results
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					

```
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Q4: From your results, what proportion of the original variance is captured by the first principal component (PC1)?

Answer Q4: The proportion of original variance captured by the first principal component (PC1) is 44.2%. *PC1 capture 0.442 of the total variance in the data, 44.2% variance.*

```
# Call our Importance table to find Proportion Variance of PC1
summary(wisc.pr)$importance[2,1]
```

```
[1] 0.44272
```

Q5: How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

Answer Q5: In order to describe at least 70% of the original variance we would need the first 3 Principal Components.

```
# Use sum() function to find the first components that describe at least 70%
# of original variance in data

sum(summary(wisc.pr)$importance[3,]<=0.70) + 1
```

```
[1] 3
```

Q6: How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Answer Q6: Nine principal components are needed in order to describe at least 90% of the original variance in the data.

```
# Use sum() function to find the first components that describe at least 70%
# of original variance in data

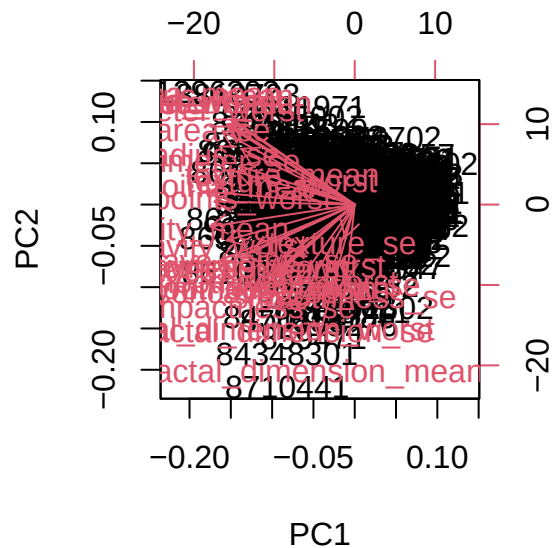
sum(summary(wisc.pr)$importance[3,]<=0.90) + 1
```

```
[1] 7
```

Interpreting PCA results

Create a biplot of the wisc.pr using the biplot() function.

```
biplot(wisc.pr)
```



Q7: What stands out to you about this plot? Is it easy or difficult to understand? Why?

Answer Q7: The thing that stands out the most is that the row names are being plotted as characters. This biplot is very difficult to understand. Since rownames are used as the plotting character in the biplot shown above, we are unable to see a clear representation of a trend.

Generate Standard Scatter Plot

```
# Call out ggplot2 from library

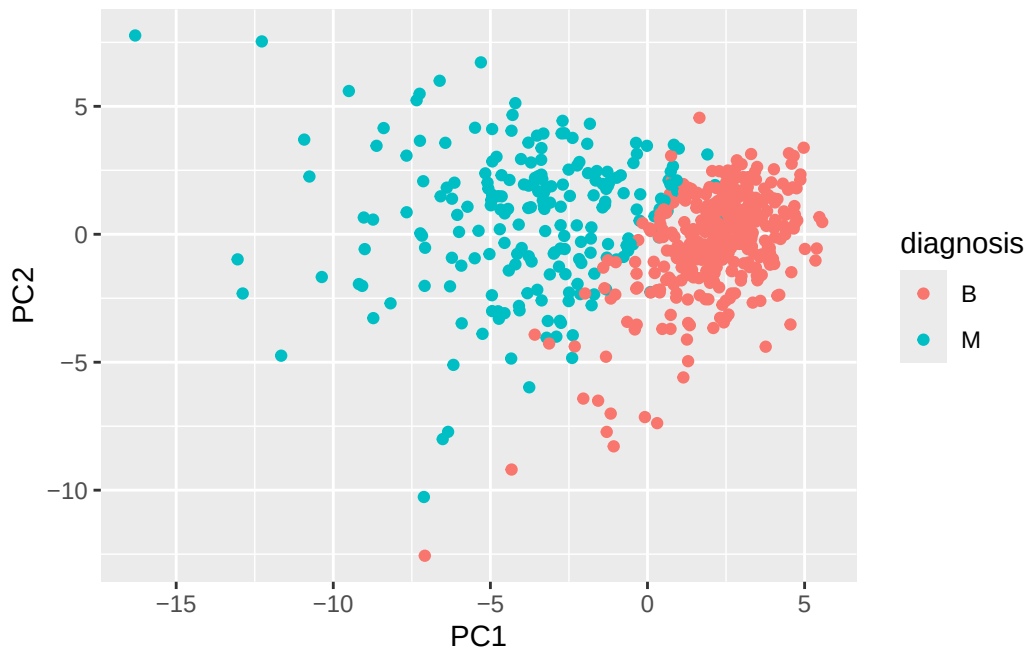
# Scatter plot observations by Components 1 & 2

library(ggplot2)

ggplot(wisc.pr$x) +
```



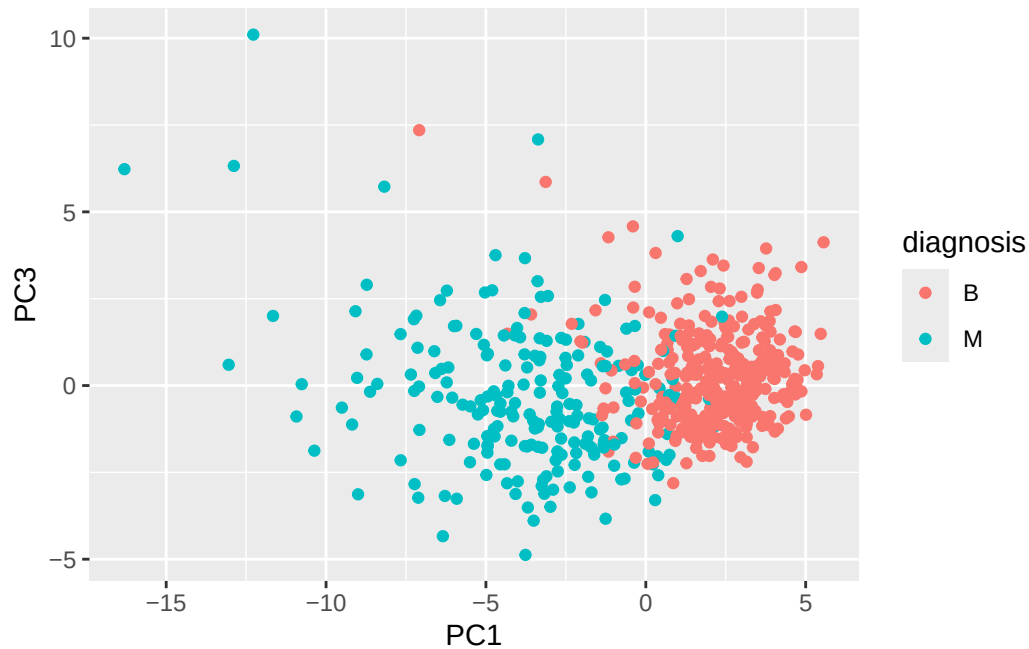
```
aes(x=PC1,y=PC2 , col=diagnosis) +  
geom_point()
```



Q8: Generates a similar plot for principal components 1 and 3. What do you notice about these plots?

Answer Q8: Shown below is the plot for PC1 and PC3 and when comparing these results to the plot above (PC1 and PC2) the separation between benign (B) and malignant (M) samples is still visible. Overall, both plots help indicate that component 1 is important for capturing a separation between the benign (B) and the malignant (M) samples. In other words, PC1 helps divide out data into 2 clusters that separate the benign and malignant sample diagnoses.

```
# Repeat for components 1 and 3  
  
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



Variance explained

Calculate Variance of components

```
# Calculate variance of each component

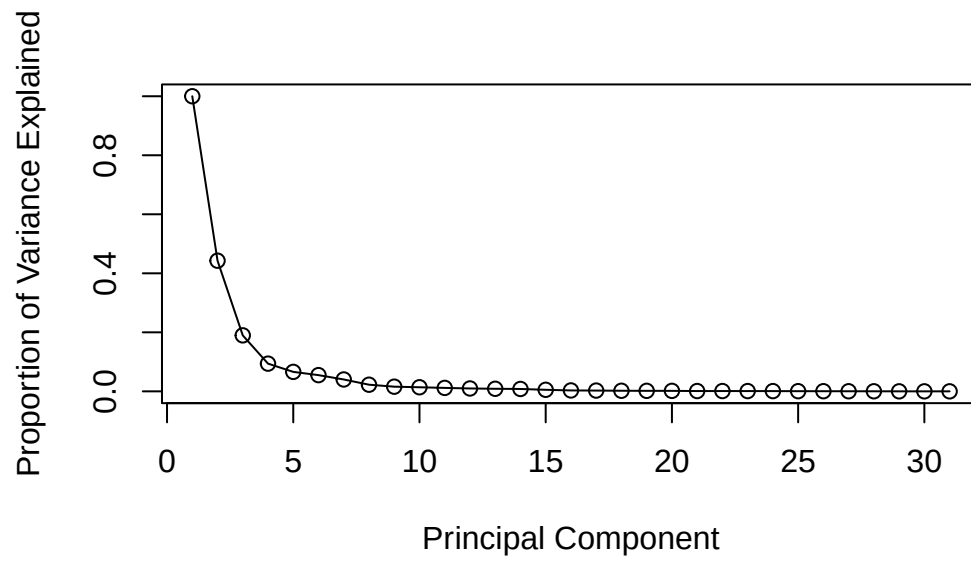
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Calculate the variance explained by: $PC\# / TotVar$

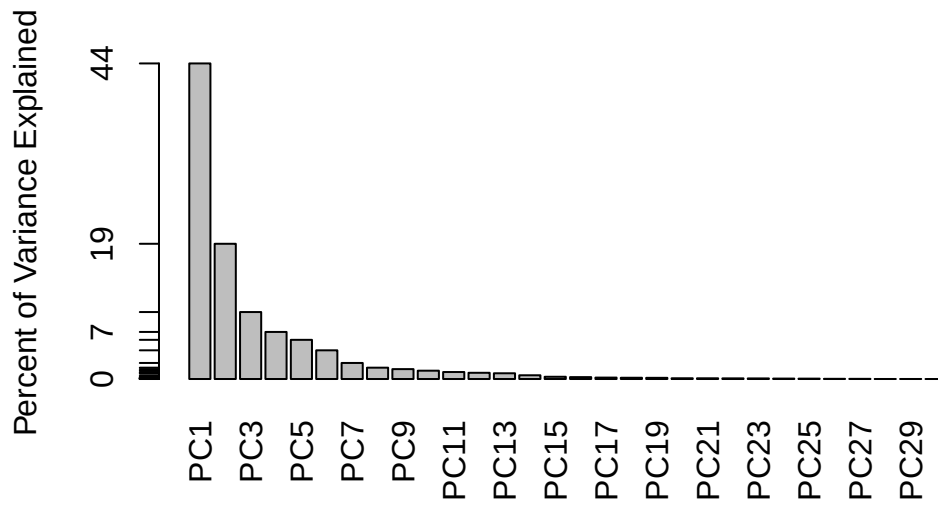
```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Alternative Histogram

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

Loading Vector (`wisc.pr$rotation`) tells us which original measurements contribute most to each Principal Component

Q9: For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

Answer Q9: The component of the loading vector for the feature “`concave.points_mean`” is -0.2608538. There are about 30 features with contributions of a larger value than that of `concave.points_mean`. Show below are the results for the component loading vector of `concave.points_mean` as well as the features with larger contributions.

```
# Loading for concave.points_mean on PC1
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
## Check features with larger contributions
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

Section 4: Hierarchical clustering

Scaling data and assigning results

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
```

Calculate Distances and assign results

```
data.dist <- dist(data.scaled)
```

Create Hierarchical Clustering

```
wisc.hclust <- hclust(data.dist, method="complete")
```

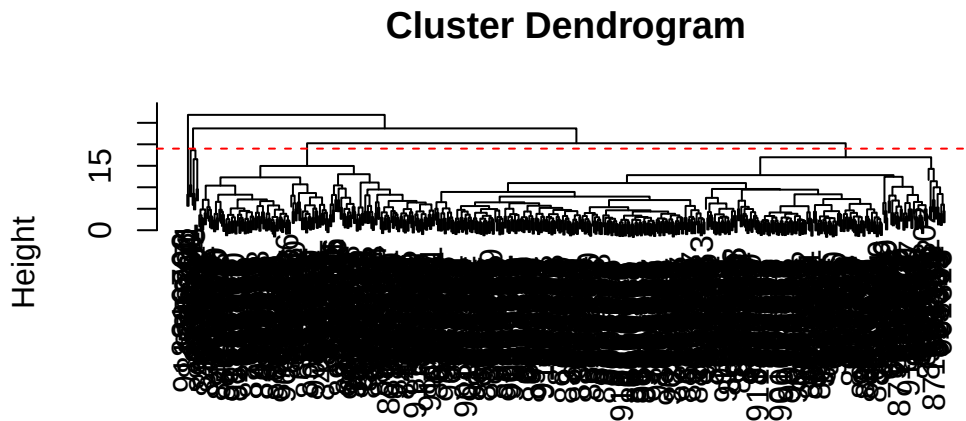
Results of hierarchical clustering

Create a Cluster Dendrogram

Q10: Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

Answer Q:10 The approximate height that results in 4 clusters is $h=19$. As shown in the R code and diagram below, past height 19 we see the cluster dendrogram section into 4 different clusters.

```
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



```
data.dist
hclust (*, "complete")
```

Selecting number of clusters

Evaluate clusters using Cutree Function

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
```

Table: compare Cluster Membership to Actual diagnosis

```
table(wisc.hclust.clusters, diagnosis)
```

```
      diagnosis  
wisc.hclust.clusters  B  M  
1    12 165  
2     2   5  
3   343  40  
4     0   2
```

Q11: OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

Answer Q11: *Skipped*

Using different methods

Different “*methods*” are used to combine points during hierarchical clustering.

Q12: Which method gives your favorite results for the same `data.dist` dataset? Explain your reasoning.

Answer Q12: My favorite methods are “complete” and “ward.D2” as these two show us the clearest models. My top favorite would be “ward.D2” because the clusters seem much more organized than any other method and the rownames (although many of them) appear to be more level. The bottom up orientation makes the graph look appealing. On the other hand, my least favorite was the average method, I understnad it diminishes the affect of outliers but it has a very large variance and makes the cluster dendrogram harder to understand.

Section 5: Combining methods

Clustering on PCA results

Combining PCA and Hierarchical Clustering - First identify the minimum number of PCs required to describe *at least 90%* of the variability in the data. Then create an H-clustering model with linkage method being “ward.D2”.

H-Cluster Dendrogram:

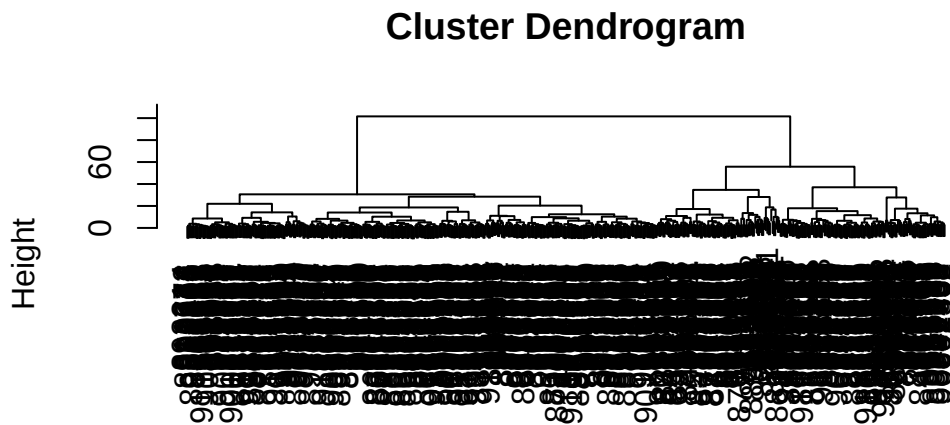
```
# How many PCs are required for 90% mark
sum(summary(wisc.pr)$importance[3,]<=0.90) + 1
```

```
[1] 7
```

```
# Combine PCA and H-cluster

wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]),
                        method = "ward.D2" )

plot(wisc.pr.hclust)
```



```
dist(wisc.pr$x[, 1:7])
hclust (*, "ward.D2")
```

*Two main branches indicate two main clusters (B? & M?)

```
# Divide the Dendrogram: 2 main clusters
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
216 353
```

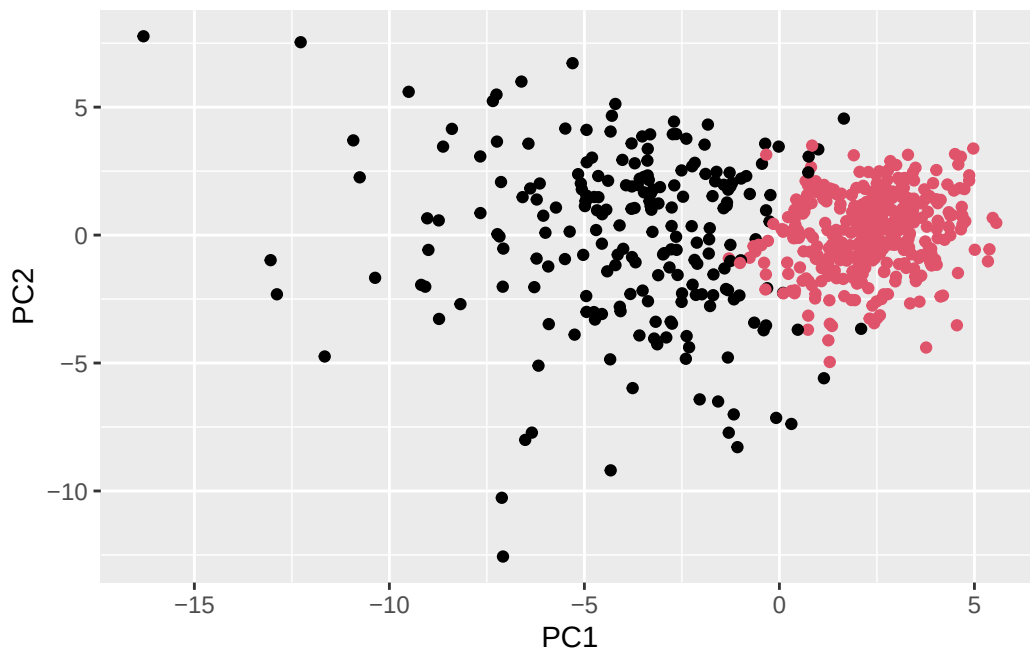


```
# Compare cluster to actual diagnosis
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
1     28 188
2    329  24
```

This indicated that cluster 1 is predominantly malignant and cluster 2 is predominantly benign.

```
# Plot PCA results colored by 2 h-cluster
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



Note: This plot looks very similar to that of the Malignant and Benign sample plot. Skipped 3D because will not work well with PDF.

Copied example from lab to answer questions 13 and 14

```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]), method="ward.D2")

wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)

table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24

```

Q13: How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

Answer Q13: This newly created h-clust model with two clusters separates out the “M” and “B” diagnoses very accurately. The majority of the “Malignant” samples are found in cluster 1 and the majority of the “Benign” clusters are found in cluster 2. Although the separation is not perfect, it is still a very well model considering this was a “blind” analysis (we did not provide the answers for B & M separation).

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24

```

Q14: How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

Answer Q14: The hierarchical clustering models created in previous section (without PCA) are not as effective in terms of separating the diagnoses. We see this in the uneven and inaccurate distributions amongst the 4 clusters. Overall the Benign samples are clustered out very well but the malignant samples seem to be spread across multiple samples. This prevents an accurate separation between the two samples: M & B. (SHOWN BELOW)

```
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B  M
1    12 165
2     2   5
3   343  40
4     0   2

```

Section 6: Sensitivity/Specificity

Q15: OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

Answer Q15: *Skipped*

Section 7: Prediction

Use predict function to take PCA model from before and NEW cancer cell data and project that NEW data onto our PCA

```

#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc

```

```

      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10     PC11     PC12     PC13     PC14
[1,] -0.2307350  0.1029569 -0.9272861  0.3411457  0.375921  0.1610764  1.187882
[2,] -0.3307423  0.5281896 -0.4855301  0.7173233 -1.185917  0.5893856  0.303029
      PC15     PC16     PC17     PC18     PC19     PC20
[1,]  0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,]  0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
      PC21     PC22     PC23     PC24     PC25     PC26
[1,]  0.1228233  0.09358453  0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776  0.01732146  0.06316631 -0.2338618 -0.20755948 -0.009833238
      PC27     PC28     PC29     PC30

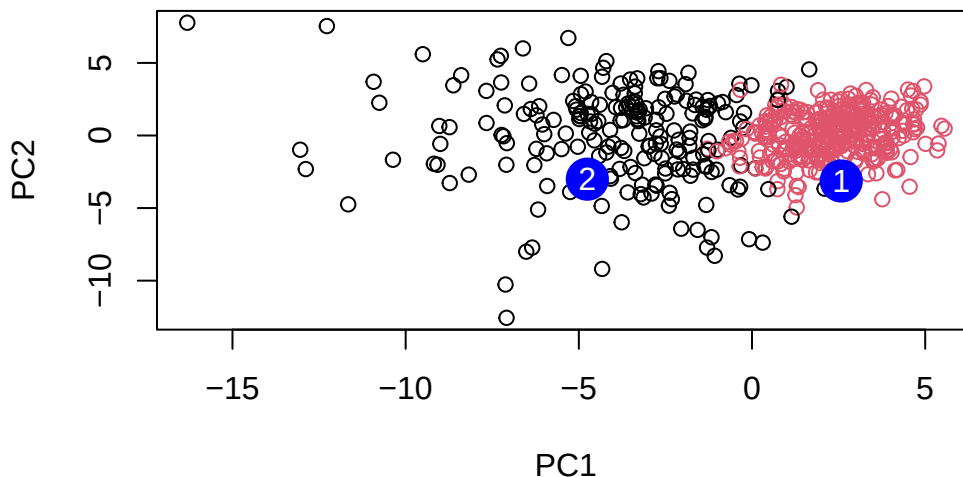
```

```
[1,] 0.220199544 -0.02946023 -0.015620933 0.005269029
[2,] -0.001134152 0.09638361 0.002795349 -0.019015820
```

This table reveals that in PC1 cluster 1 (which is predominantly malignant as stated before) has a much larger value than cluster 2. This means that cluster 1 will be found more to the right and cluster 2 will be towards the left on this plot.

Plot New Data

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16: Which of these new patients should we prioritize for follow up based on your results?

Answer Q16: Based on these results, we should prioritize the follow up for Patient 1. When these new patients were overlayed onto the existing PCA space, we see that patient 1 fell in cluster group 1 which has been identified as predominantly malignant. On the other hand, patient 2 falls in cluster 2 which is predominantly benign. Patient 1's location on the (predominantly) "Malignant" cluster indicates that they are at higher risk and that they require immediate medical attention; therefore, Patient 1 should be prioritized for a follow up consultation.